Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala

Computer Science & Engineering Department

(February 20, 2026); 5:30PM-8:30PM

BE-Computer Engineering, Auxiliary Exam

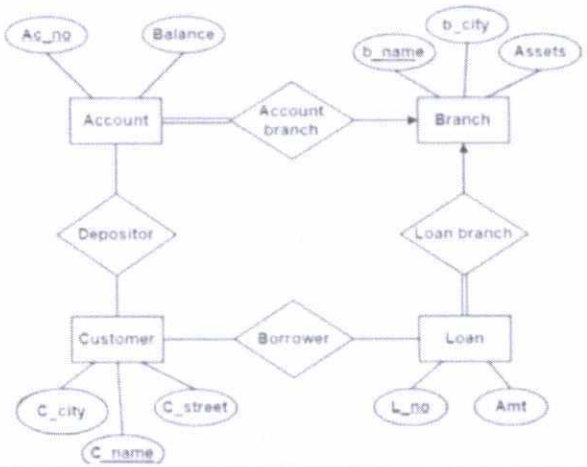
Course Code: **UCS310**

Course Name: **Database Management System**

Time: **3 Hours**, M. Marks: **100**

Name of Faculty: Dr. Ananya Pandey

Note: *Attempt any FIVE questions. All questions carry equal marks.*
Provide proper justification and neat diagrams wherever required. Assume suitable missing data, if any.

Q. No	Questions	M.M.
1.	Explain the following with suitable examples: (a) Two-phase locking protocol *(5 Marks)* (b) Concurrency control mechanism *(5 Marks)* (c) 3NF and BCNF *(5 Marks)* (d) Generalization and Specialization *(5 Marks)*	20
2.	(a) Explain the Entity–Relationship (ER) model in detail. Discuss its components such as entities, attributes, relationships, cardinality constraints, and participation constraints. Illustrate your answer with suitable examples. *(10 Marks)* (b) Determine the minimum number of tables required when converting the given ER diagram into a relational schema. List all the attributes of each table. *(10 Marks)* 	20
3.	(a) Consider the following database schema for a university system. Formulate the SQL statements for the following queries. Student: Stores information about students. Student (StudentED, Name, Age, Major) Course: Stores information about courses. Course (CourseID, CourseName, Credits) Enrolled: Tracks which students are enrolled in which courses. Enrolled (StudentID, CourseID, Grade) Professor: Stores information about professors. Professor (ProfessorID, Name, Department) Teaches: Tracks which professors teach which courses. Teaches (ProfessorID, CourseID) 1. Find the names of students who have taken at least one course with a grade of "A". 2. Retrieve the list of all students who are not enrolled in any course.	20

3. Retrieve the names of professors who teach courses taken by students majoring in "Mathematics."
4. List all professors who do not teach any courses.
5. Find the average age of students enrolled in courses offered by the 'Computer Science' department. *(10 Marks)*

(b) Explain the Structured Query Language (SQL) in detail. Discuss its different categories such as DDL, DML, DCL, and TCL with suitable examples. *(10 Marks)*

4. (a) Consider the relation R (P, Q, S, T, X, Y, Z, W) with the following FDs:
PQ→X
P→YX
Q→Y
Y→ZW
Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes:
D1: R = [(P, Q, S, T); (P, T, X); (Q, Y); (Y, Z, W)]
D2: R = [(P, Q, S); (T, X); (Q, Y); (Y, Z, W)]
Check whether D1 and D2 are Lossy or Lossless Join Decomposition. *(10 Marks)*

(b) What are subqueries in SQL? Explain different types of subqueries and their uses. How do correlated subqueries differ from non-correlated subqueries? *(10 Marks)*

20

5. (a) Consider the schedule and the following scenario:
- Transaction T1: Assigned TS(T1) =5
- Transaction T2: Assigned TS(T2) =10
- Initial states of data items:

a. X: RTS(X)=0, WTS(X)=0
b. Y: RTS(Y)=0, WTS(Y)=0

T_1	T_2
READ (X)	
	WRITE (X)
WRITE (Y)	
	READ (Y)

Apply the Timestamp Ordering Protocol to the above schedule. Determine whether each operation is executed or rejected. Show the updated values of RTS and WTS after each step and state whether any transaction will be rolled back. (10 marks)

(b) Consider the following schedule of transactions:

Time	Transactions	Operations	Descriptions
T_1	T_1	READ (A)	T_1 reads the value of A
T_2	T_2	READ (A)	T_2 reads the value of A
T_3	T_1	WRITE (A = A + 10)	T_1 updates the value of A
T_4	T_2	WRITE (A = A * 2)	T_2 updates the value of A
T_5	T_2	COMMIT	T_2 commits the changes
T_6	T_1	COMMIT	T_1 commits the changes

1. Is this schedule **recoverable**?
2. If not, identify the issue and suggest how the schedule can be made recoverable. (10 marks)

20

6.	(a) For the given schedule, determine whether it is conflict serializable and view serializable. If the schedule is serializable, identify the equivalent serial schedule. *(10 Marks)* (b) Consider the following schedule involving two transactions T_1 and T_2. Does the schedule follow the two-phase locking (2PL) protocol? Justify your answer. Identify whether the schedule results in a deadlock. If yes, explain how it occurs. *(10 Marks)*	20

T1	T2	T3
READ (A)		
	READ (A)	
WRITE (A)		
		READ (A)
	WRITE (A)	
		WRITE (B)
WRITE (B)		

T_1	T_2
LOCK – X (A)	
READ (A)	
	LOCK – S (B)
	READ (B)
LOCK – X (B)	
WRITE (B)	
UNLOCK (A)	
UNLOCK (B)	
	LOCK – S(A)
	READ (A)
	UNLOCK (B)